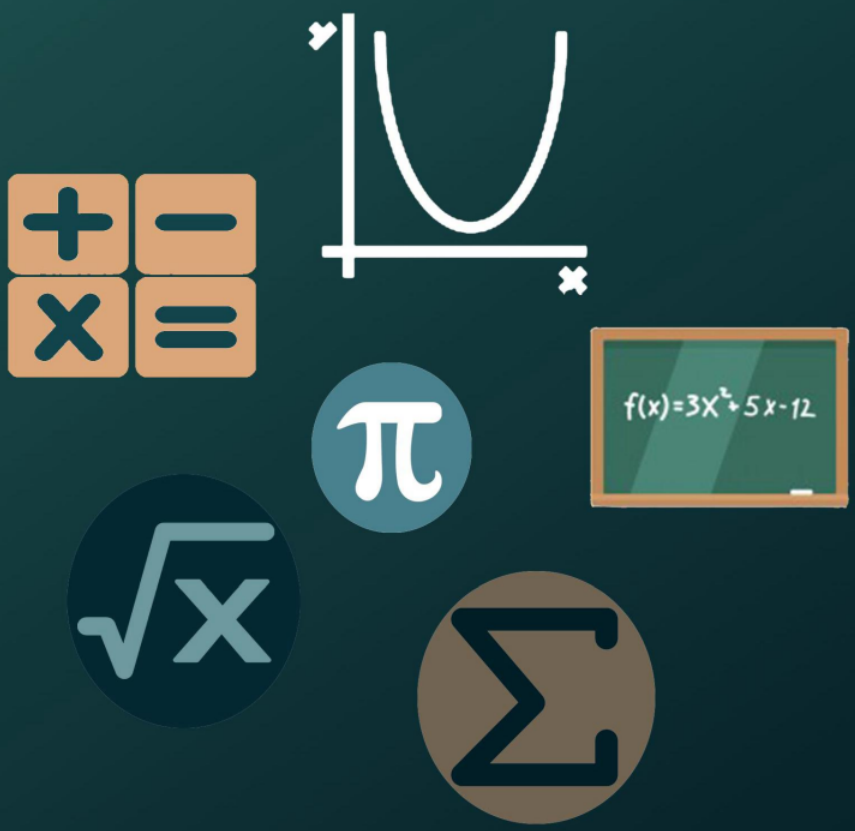
MATHEMATICS

Hyperbola



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Standard Hyperbola	1 - 12	1, 12 - 13	1,2	
2	Hyperbola at a Glance				
3	Other Forms of Hyperbola	13 - 15	2, 14	3,4	1 - 2
4	Auxilliary Circle and Eccentricity of Hyperbola	16,17		5,6	
5	Point and Hyperbola				
6	Tangents to Hyperbola - Slope Form	18-20	3 - 4, 15		3 - 5, 19
7	Tangents to Hyperbola - Parametric Form	21		7	
8	Tangents to Hyperbola - Cartesian Form and Chords	22,23	5	8	16
9	Normals to Hyperbola	24,25			6 - 7, 20
10	Pair of Tangents & Chord with given Mid-Point	26,27		9-14	8, 21
11	Asymptotes	28	6	15	9
12	Rectangular Hyperbola	29-32	7 - 10, 16	16-21	10 - 12
13	After Complete Chapter		11	22	13 - 15, 17 - 18, 22 - 25

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Standard Hyperbola

Q1. If the centre, vertex and focus of a hyperbola be (0, 0), (4, 0) and (6, 0) respectively, then the equation of the hyperbola is-

(A) $4x^2 - 5y^2 = 8$

(B) $4x^2 - 5y^2 = 80$

(C) $5x^2 - 4y^2 = 80$

(D) $5x^2 - 4y^2 = 8$

Q2. The foci of the hyperbola $4x^2 - 9y^2 - 36 = 0$ are

(A) $\left(\pm\sqrt{11}, 0\right)$ (B) $\left(\pm\sqrt{12}, 0\right)$

(C) $\left(\pm\sqrt{13}, 0\right)$ (D) $\left(0, \pm\sqrt{12}\right)$

Q3. The length of the transvere axis of a hyperbola is 7 and it passes through the point (5, – 2). The equation of the hyperbola is-

(A) $\frac{4}{49}x^2 - \frac{196}{51}y^2 = 1$

(B) $\frac{49}{196}x^2 - \frac{51}{196}y^2 = 1$

(C) $\frac{4}{49}x^2 - \frac{51}{196}y^2 = 1$

(D) None of these

Q4. The vertices of a hyperbola are at (0, 0) and (10, 0) and one of its foci is at (18, 0). The equation of the hyperbola is -

(A) $\frac{x^2}{25} - \frac{y^2}{144} = 1$

(B) $\frac{(x-5)^2}{25} - \frac{y^2}{144} = 1$

(C) $\frac{x^2}{25} - \frac{(y-5)^2}{144} = 1$

(D) $\frac{(x-5)^2}{25} - \frac{(y-5)^2}{144} = 1$

Q5. The equation $\frac{x^2}{12-\lambda} + \frac{y^2}{8-\lambda} = 1$ represents-

(A) a hyperbola if $\lambda < 8$

(B) an ellipse if $\lambda > 0$

(C) a hyperbola if $8 < \lambda < 12$

(D) None of these

Q6. The latus rectum of the hyperbola $16x^2 - 9y^2 = 144$ is-

(A) 16/3 (B) 32/3

(C) 8/3 (D) 4/3

Q7. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide.

Then the value of b^2 is-

(A) 9 (B) 1

(C) 5 (D) 7

Q8. The distance between the directrices of a rectangular hyperbola is 10 unit, then distance between its foci is :

(A) $10\sqrt{2}$ (B) 5

(C) $5\sqrt{2}$ (D) 20

Q9. $(x-1)^2 + (y-2)^2 = \frac{3(2x+3y+2)^2}{13}$

represents hyperbola whose eccentricity is-

(A) $\frac{\sqrt{13}}{\sqrt{3}}$ (B) $\frac{3}{\sqrt{13}}$

(C) $\sqrt{3}$ (D) 3

Q10. Parametric form of the hyperbola $\frac{x^2}{4}-\frac{y^2}{9}=-1$ is -

(A) $(2\tan\theta, 3\sec\theta)$ (B) $(3\sec\theta, 2\tan\theta)$

(C) $(9\sec\theta, 4\tan\theta)$ (D) None of these

Q11. Locus of the point of intersection of straight lines $\frac{x}{a}-\frac{y}{b}=m$ and $\frac{x}{a}+\frac{y}{b}=\frac{1}{m}$ is :

(A) An ellipse (B) A circle

(C) A hyperbola (D) A parabola

Q12. For hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$ which of the following remains constant with change in α -

(A) abscissae of vertices

(B) abscissae of foci

(C) eccentricity

(D) directrix

Other Forms of Hyperbola

Q13. If e and e' be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e^2}+\frac{1}{e'^2}=$

(A) 0 (B) 1

(C) 2 (D) None of these

Q14. The equation of the transverse and conjugate axis of the hyperbola $16x^2-y^2+64x+4y+44=0$ are-

(A) $x=2, y+2=0$

(B) $x=2, y=2$

(C) $y=2, x+2=0$

(D) None of these

Q15. The focal length of the hyperbola $x^2-3y^2-4x-6y-11=0$, is-

(A) 4 (B) 6

(C) 8 (D) 10

Q16. If the eccentricity of the hyperbola $x^2-y^2\sec^2\alpha=5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\alpha+y^2=25$, then a value of α is :

(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\pi/2$

Q17. Eccentricity of the hyperbola conjugate to the hyperbola $\frac{x^2}{4}-\frac{y^2}{12}=1$ is -

(A) $\frac{2}{\sqrt{3}}$ (B) 2 (C) $\sqrt{3}$ (D) $\frac{4}{3}$

Q18. Line $\ell x + my + n = 0$ is a tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if-

(A) $a^2\ell^2 - b^2m^2 = n^2$

(B) $a^2\ell^2 + m^2 = n^2b^2$

(C) $a^2 + b^2 = n^2(\ell^2 + m^2)$

(D) $a^2\ell^2 + b^2m^2 = n^2$

Q19. The value of m for which $y = mx + 6$ is a tangent to the hyperbola $\frac{x^2}{100}-\frac{y^2}{49}=1$ is -

(A) $\sqrt{\frac{17}{20}}$ (B) $\sqrt{\frac{20}{17}}$

(C) $\sqrt{\frac{3}{20}}$ (D) $\sqrt{\frac{3}{20}}$

Q20. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is -

(A) a hyperbola (B) a parabola

(C) a circle (D) an ellipse

Q21. If the tangent on the point $(2\sec\phi, 3\tan\phi)$ of the hyperbola $\frac{x^2}{4}-\frac{y^2}{9}=1$ is parallel to $3x-y+4=0$, then the value of ϕ is :

(A) 75° (B) 45° (C) 60° (D) 30°

Q22. The point of contact of the line $2x+\sqrt{6}y=2$ and the hyperbola $x^2-2y^2=4$ is -

(A) $(4, -\sqrt{6})$ (B) $(\sqrt{6}, 1)$

(C) $\left(\frac{1}{2}, \frac{1}{\sqrt{6}}\right)$ (D) $\left(\frac{1}{6}, \frac{3}{2}\right)$

Q23. The magnitude of the gradient of the tangent at an extremity of latus rectum of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is equal to (where e is the eccentricity of the hyperbola).

(A) be (B) e

(C) ab (D) ae

Q24. In which of the following cases maximum number of normals can be drawn from a point P lying in the same plane.

(A) circle (B) parabola

(C) ellipse (D) hyperbola

Q25. The equation of the normal to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ at the point $(8, 3\sqrt{3})$ is :

(A) $\sqrt{3}x+2y=25$ (B) $2x+\sqrt{3}y=25$

(C) $y+2x=25$ (D) $x+y=25$

Q26. The tangents from $(1, 2\sqrt{2})$ to the hyperbola $16x^2-25y^2=400$ include an angle between them =

(A) $\pi/6$ (B) $\pi/4$

(C) $\pi/3$ (D) $\pi/2$

Q27. The number of points from where a pair of perpendicular tangents can be drawn to the hyperbola, $x^2\sec^2\alpha - y^2\operatorname{cosec}^2\alpha = 1$, $\alpha \in (0, \pi/4)$ are -

(A) 0 (B) 1

(C) 2 (D) infinite

Q28. The combined equation of the asymptotes of the hyperbola $2x^2+5xy+2y^2+4x+5y=0$ is :

(A) $2x^2+5xy+2y^2=0$

(B) $2x^2+5xy+2y^2-4x+5y+2=0$

(C) $2x^2+5xy+2y^2+4x+5y-2=0$

(D) $2x^2+5xy+2y^2+4x+5y+2=0$

Q29. If $(a-2)x^2+ay^2=4$ represents rectangular hyperbola, then a equals -

(A) 0 (B) 2 (C) 1 (D) 3

Q30. Consider the set of hyperbola $xy=k$, $k\in R$. Let e_1 be the eccentricity when $k=4$ and e_2 be the eccentricity when $k=9$, then e_1-e_2 is equal to -

(A) –1 (B) 0 (C) 2 (D) 3

Q31. The equation to the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy=c^2$ is :

(A) $\frac{x}{x_1+x_2}+\frac{y}{y_1+y_2}=1$

(B) $\frac{x}{x_1-x_2}+\frac{y}{y_1-y_2}=1$

(C) $\frac{x}{y_1+y_2}+\frac{y}{x_1+x_2}=1$

(D) $\frac{x}{y_1-y_2}+\frac{y}{x_1-x_2}=1$

Q32. For the rectangular hyperbola $xy=\frac{a^2}{2}$ the co-ordinates of its foci are -

(A) $(\sqrt{2}a, \sqrt{2}a), (-\sqrt{2}a, -\sqrt{2}a)$

(B) $(\sqrt{2}a, -\sqrt{2}a), (-\sqrt{2}a, \sqrt{2}a)$

(C) (a, a), (–a, –a)

(D) $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)\left(-\frac{a}{\sqrt{2}}, -\frac{a}{\sqrt{2}}\right)$

Exercise - 1A
Concept Builder

ONE OR MORE THAN

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The equation $\frac{x^2}{29-p}+\frac{y^2}{4-p}=1$ $(p \neq 4, 29)$ represents -

(A) an ellipse if p is any constant greater than 4

(B) a hyperbola if p is any constant between 4 and 29.

(C) a rectangular hyperbola if p is any constant greater than 29.

(D) no real curve is p is greater than 29.

Q2. Let p and q be non-zero real numbers.

Then the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents

(A) two straight lines and a circle, when $r = 0$ and p, q are of the opposite sign.

(B) two circles, when $p = q$ and r is of sign opposite to that of p.

(C) a hyperbola and a circle, when p and q are of opposite sign and $r \neq 0$.

(D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p.

Q3. Equations of a common tangent to the two hyperbolas $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, and $\frac{y^2}{a^2}-\frac{x^2}{b^2}=1$ is-

(A) $y = x + \sqrt{a^2 - b^2}$

(B) $y = x - \sqrt{a^2 - b^2}$

(C) $y = -x + \sqrt{a^2 - b^2}$

(D) $y = -x - \sqrt{a^2 - b^2}$

Q4. The equation of a tangent to the hyperbola $16x^2 - 25y^2 - 96x + 100y - 356 = 0$ which makes an angle $\pi/4$ with the transverse axis is -

(A) $y = x + 2$ (B) $y = x - 4$

(C) $x = y + 3$ (D) $x = y + 2$

Q5. Locus of the feet of the perpendiculars drawn from either foci on a variable tangent to the hyperbola $16y^2 - 9x^2 = 1$ is -

(A) $x^2 + y^2 = 9$ (B) $x^2 + y^2 = 1/9$

(C) $x^2 + y^2 = 7/144$ (D) $x^2 + y^2 = 1/16$

Q6. If $H(x, y) = 0$ represent the equation of a hyperbola and $A(x, y) = 0$, $C(x, y) = 0$ the equations of its asymptotes and the conjugate hyperbola respectively then for any point (α, β) in the plane : $H(\alpha, \beta)$, $A(\alpha, \beta)$ and $C(\alpha, \beta)$ are in -

(A) A.P. (B) G.P.

(C) H.P. (D) None

Q7. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1,y_1)$, $Q(x_2,y_2)$, $R(x_3,y_3)$, $S(x_4,y_4)$, then -

(A) $x_1 + x_2 + x_3 + x_4 = 0$

(B) $y_1 + y_2 + y_3 + y_4 = 0$

(C) $x_1x_2x_3x_4 = c^4$

(D) $y_1y_2y_3y_4 = c^4$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer. For each question, choose the options corresponding to the correct answers.*

Paragraph # 1 (Ques. 8 to 10.)

The graph of the conic $x^2 - (y - 1)^2 = 1$ has one tangent line with positive slope that passes through the origin. the point of tangency being (a, b). Then-

Q8. The value of $\sin^{-1}\left(\frac{a}{b}\right)$ is

(A) $\frac{5\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q9. Length of the latus rectum of the conic is

(A) 1 (B) $\sqrt{2}$ (C) 2 (D) None

Q10. Eccentricity of the conic is

(A) $\frac{4}{3}$ (B) $\sqrt{3}$

(C) 2 (D) $\sqrt{2}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11.

Column I	**Column II**
(A) The area of the triangle that a tangent at a point of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ makes with its asymptotes	(P) 12
(B) If the line $y = 3x + \lambda$ touches the curve $9x^2 - 5y^2 = 45$ then $\lvert\lambda\rvert$ is	(Q) 6
(C) If the chord $x\cos\alpha + y\sin\alpha = P$ of the hyperbola $\frac{x^2}{16}-\frac{y^2}{18}=1$ subtends a right angle at the centre, then the radius of the circle concentric with the hyperbola to which the given chord is a tangent is	(R) 24
(D) If λ be the length of the latus rectum of the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$, then 3λ is equal to	(S) 32

SUBJECTIVE TYPE

Q12. Find the equation to the hyperbola of given transverse axis whose vertex bisects the distance between the centre and the focus.

Q13. Find the equation of the hyperbola whose directrix is $2x + y = 1$, focus $(1, 2)$ and eccentricity $\sqrt{3}$.

Q14. Consider the hyperbola
$9x^2 - 16y^2 + 72x - 32y - 16 = 0$.
Find the following :

(A) centre
(B) eccentricity
(C) focii
(D) equation of directrix
(E) length of the latus rectum
(F) equation of auxilary circle
(G) equation of director circle

Q15. The equation of a tangent to the hyperbola $3x^2 - y^2 = 3$, parallel to the line $y = 2x + 4$ is

Q16. Find the eccentricity of the conic represented by $x^2 - y^2 - 4x + 4y + 16 = 0$.

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. With one focus of the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$ as the centre, a circle is drawn which is tangent to the hyperbola with no part of the circle being outside the hyperbola. The radius of the circle is -

(A) less than 2 (B) 2

(C) $\frac{11}{3}$ (D) none

Q2. The locus of the point of intersection of the line $\sqrt{3}x-y-4\sqrt{3}k=0$ and $\sqrt{3}kx+ky-4\sqrt{3}=0$ is a hyperbola of eccentricity.

(A) 1 (B) 2

(C) 2.5 (D) $\sqrt{3}$

Q3. A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is -

(A) $x^2\operatorname{cosec}^2\theta - y^2\sec^2\theta = 1$

(B) $x^2\sec^2\theta - y^2\operatorname{cosec}^2\theta = 1$

(C) $x^2\sin^2\theta - y^2\cos^2\theta = 1$

(D) $x^2\cos^2\theta - y^2\sin^2\theta = 1$

Q4. Consider a branch of the hyperbola $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A, then the area of the triangle ABC is -

(A) $1-\sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{3}{2}}-1$

(C) $1+\sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{3}{2}}+1$

Q5. The area of the quadrilateral with its vertices at the foci of the conics

$9x^2-16y^2-18x+32y-23=0$ and

$25x^2 + 9y^2 - 50x - 18y + 33 = 0$, is -

(A) 5/6 (B) 8/9

(C) 5/3 (D) 16/9

Q6. At the point $P(1,\sqrt{3})$ on the circle $x^2+y^2=4$ a tangent is drawn which meets the transverse axis of the hyperbola $\frac{x^2}{4}-\frac{y^2}{1}=1$ at Q. From Q a line is drawn parallel to conjugate axis, which cuts the hyperbola at R above the x-axis, then PR equals -

(A) 3 (B) $\sqrt{10}$

(C) $\frac{3\sqrt{3}+4}{2}$ (D) None of these

Q7. If a variable straight line $x\cos\alpha + y\sin\alpha = p$, which is a chord of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1(b>a)$, subtend a right angle at the centre of the hyperbola, then it always touches a fixed circle whose radius is

(A) $\frac{ab}{\sqrt{(b-2a)}}$ (B) $\frac{a}{\sqrt{(a-b)}}$

(C) $\frac{ab}{\sqrt{(b^2-a^2)}}$ (D) $\frac{ab}{b\sqrt{(b+a)}}$

Q8. The condition that a straight line with slope m will be normal to parabola $y^2 = 4ax$ as well as a tangent to rectangular hyperbola $x^2 - y^2 = a^2$ is -

(A) $m^6-4m^2+2m-1=0$

(B) $m^4+3m^3+2m+1=0$

(C) $m^6-2m=0$

(D) $m^6+4m^4+3m^2+1=0$

Q9. Locus of the point of intersection of the tangents at the points with eccentric angles ϕ and $\frac{\pi}{2}-\phi$ on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is :

(A) $x = a$ (B) $y = b$

(C) $x = ab$ (D) $y = ab$

Q10. Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to 2R and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola then the correct relation is -

(A) $4e_1^2 - e_2^2 = 6$ (B) $e_1^2 - 4e_2^2 = 2$

(C) $4e_2^2 - e_1^2 = 6$ (D) $2e_1^2 - e_2^2 = 4$

Q11. The number of possible tangents which can be drawn to the curve $4x^2 - 9y^2 = 36$, which are perpendicular to the straight line $5x + 2y - 10 = 0$ is :

(A) zero (B) 1 (C) 2 (D) 4

Q12. The equation of the common tangent to the curve $y^2 = 8x$ and $xy = -1$ is -

(A) $3y = 9x + 2$ (B) $y = 2x + 1$

(C) $2y = x + 8$ (D) $y = x + 2$

Q13. A tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with centre C meets its director circle at P and Q. Then the product of the slopes of CP and CQ, is -

(A) $\frac{9}{4}$ (B) $\frac{-4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$

Q14. The locus of the mid-point of the chords of the hyperbola $(x^2/a^2) - (y^2/b^2) = 1$ passing through a fixed point (α, β) is

(A) $(x - \alpha/2)^2/a^2 - (y - \beta/2)^2/b^2 = \alpha^2/4a^2 - \beta^2/4b^2$

(B) $(x + \alpha/2)^2/a^2 - (y - \beta/2)^2/b^2 = \alpha^2/4a^2 + \beta^2/4b^2$

(C) $(x - \alpha/2)^2/a^2 - (y - \beta/2)^2/b^2 = \alpha^2/4a^2 + \beta^2/4b^2$

(D) None of these

Q15. The asymptote of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ form with any tangent to the hyperbola a triangle whose area is $a^2\tan\lambda$ in magnitude then its eccentricity is :

(A) $\sec\lambda$ (B) $\operatorname{cosec}\lambda$

(C) $\sec^2\lambda$ (D) $\operatorname{cosec}^2\lambda$

Q16. For each positive integer n, consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$ then $\lim_{n\to\infty}(n.d_n)$ has the value equal to-

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) 0

Q17. The chord PQ of the rectangular hyperbola $xy = a^2$ meets the axis of x at A ; C is the mid point of PQ & 'O' is the origin. Then the Δ ACO is :

(A) equilateral (B) isosceles

(C) right angled (D) right isosceles.

Q18. The tangent to the hyperbola $xy = c^2$ at the point P intersects the x-axis at T and the y-axis at T′. The normal to the hyperbola at P intersects the x-axis at N and the y-axis at N′. The areas of the triangles PNT and PN'T' are Δ and Δ' respectively, then $\frac{1}{\Delta}+\frac{1}{\Delta'}$ is -

(A) equal to 1 (B) depends on t

(C) depends on c (D) equal to 2

Q19. At the point of intersection of the rectangular hyperbola $xy = c^2$ and the parabola $y^2 = 4ax$ tangents to the rectangular hyperbola and the parabola make an angle θ and ϕ respectively with the axis of X, then -

(A) $\theta = \tan^{-1}(-2\tan\phi)$

(B) $\phi = \tan^{-1}(-2\tan\theta)$

(C) $\theta = \frac{1}{2}\tan^{-1}(-\tan\phi)$

(D) $\phi = \frac{1}{2}\tan^{-1}(-\tan\theta)$

Q20. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is -

(A) $y + mx = 0$ (B) $y - mx = 0$

(C) $my - x = 0$ (D) $my + x = 0$

Q21. If the normal to the rectangular hyperbola $xy = c^2$ at the point 't' meets the curve again at 't_1' then $t^3 t_1$ has the value equal to -

(A) 1 (B) – 1 (C) 0 (D) None

Q22. P is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T . If O is the centre of the hyperbola, then OT.ON is equal to :

(A) e^2 (B) a^2

(C) b^2 (D)b^2/a^2

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A hyperbola passes through the point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b)$ whose eccentric angle is $\frac{\pi}{4}$. If the hyperbola and the ellipse have the same foci, then the equation of hyperbola is -

(A) $x^2 - y^2 = a^2 - b^2$

(B) $x^2 - y^2 = \frac{1}{2}(a^2 - b^2)$

(C) $x^2 - y^2 = \frac{1}{4}(a^2 - b^2)$

(D) $x^2 - y^2 = \frac{1}{\sqrt{2}}(a^2 - b^2)$

Q2. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.

(A) $x = \frac{a}{2}\left(t + \frac{1}{t}\right)$ & $y = \frac{b}{2}\left(t - \frac{1}{t}\right)$

(B) $\frac{tx}{a} - \frac{y}{b} + t = 0$ & $\frac{x}{a} + \frac{ty}{b} - 1 = 0$

(C) $x = e^t + e^{-t}$ & $y = e^t - e^{-t}$

(D) $x^2 - 6 = 2\cos t$ & $y^2 + 2 = 4\cos^2\frac{t}{2}$

Q3. An ellipse has eccentricity 1/2 and one focus at the point P (1/2, 1). Its one directrix nearer to the point P is the common tangent to the circle $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$. The equation of the ellipse in standard form is -

(A) $9x^2 + 12y^2 = 108$

(B) $9(x - 1/3)^2 + 12(y - 1)^2 = 1$

(C) $9(x - 1/3)^2 + 4(y - 1)^2 = 36$

(D) None of these

Q4. A straight line touches the rectangular hyperbola $9x^2 - 9y^2 = 8$ and the parabola $y^2 = 32x$. The equation of the line is -

(A) $9x + 3y - 8 = 0$ (B) $9x - 3y + 8 = 0$

(C) $9x + 3y + 8 = 0$ (D) $9x - 3y - 8 = 0$

Q5. The line $y = x + 5$ touches.

(A) the parabola $y^2 = 20x$

(B) the ellipse $9x^2 + 16y^2 = 144$

(C) the hyperbola $\frac{x^2}{29} - \frac{y^2}{4} = 1$

(D) the circle $x^2 + y^2 = 25$

Q6. The normal at P to a hyperbola of eccentricity e, intersects its transverse and conjugate axes at L and M respectively. If locus of the mid point of LM is hyperbola, then eccentricity of the hyperbola is -

(A) $\left(\frac{e+1}{e-1}\right)$ (B) $\frac{e}{\sqrt{(e^2-1)}}$

(C) e (D) None of these

Q7. If the normal at P to the rectangular hyperbola $x^2 - y^2 = 4$ meets the axes in G and g and C is the centre of the hyperbola, then

(A) PG = PC (B) Pg = PC

(C) PG = Pg (D) Gg = 2 PC

Q8. If θ is eliminated from the equations a sec θ – x tan θ = y and b sec θ + y tan θ = x (a and b are constant) then the eliminant denotes the equation of -

(A) the director circle of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(B) auxiliary circle of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(C) Director circle of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(D) Director circle of the circle

$$x^2 + y^2 = \frac{a^2 + b^2}{2}.$$

Q9. The coordinates of a point common to a directrix and an asymptote of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ are -

(A) $\left(\frac{25}{\sqrt{41}}, \frac{20}{\sqrt{41}}\right)$ (B) $\left(-\frac{25}{\sqrt{41}}, -\frac{20}{\sqrt{41}}\right)$

(C) $\left(\frac{25}{3}, \frac{20}{3}\right)$ (D) $\left(-\frac{25}{3}, -\frac{20}{3}\right)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 10 to 12)

A conic C passes through the point (2, 4) and is such that the segment of any of its tangents at any point contained between the co-ordinate axes is bisected at the point of tangency. Let S denotes circle described on the foci F_1 and F_2 of the conic C as diameter.

Q10. Vertex of the conic C is -

(A) (2, 2), (–2, – 2)

(B) $\left(2\sqrt{2}, 2\sqrt{2}\right), \left(-2\sqrt{2}, -2\sqrt{2}\right)$

(C) (4, 4), (–4, – 4)

(D) $\left(\sqrt{2}, \sqrt{2}\right), \left(-\sqrt{2}, -\sqrt{2}\right)$

Q11. Director circle of the conic is -

(A) $x^2 + y^2 = 4$ (B) $x^2 + y^2 = 8$

(C) $x^2 + y^2 = 2$ (D) None

Q12. Equation of the circle S is -

(A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$

(C) $x^2 + y^2 = 32$ (D) $x^2 + y^2 = 4$

Paragraph # 2 (Ques. 13 to 15)

Consider the conic C : $\frac{x^2}{16} + \frac{y^2}{12} = 1$

Q13. Equation of circle touching C at one extremity of latus-rectum and passing through centre of C is/are-

(A) $8x^2 + 8y^2 - 19x - 22y = 0$

(B) $8x^2 + 8y^2 - 19x + 22y = 0$

(C) $8x^2 + 8y^2 + 19x - 22y = 0$

(D) $8x^2 + 8y^2 + 19x + 22y = 0$

Q14. The equation of parabolas with same latus-rectum as conic C, is/are-

(A) $y^2 - 6x + 3 = 0$ (B) $y^2 + 6x - 21 = 0$

(C) $y^2 - 6x - 21 = 0$ (D) $y^2 + 6x + 3 = 0$

Q15. If a hyperbola passes through extremities of minor axis of conic C and its transverse and conjugate axis coincides with the minor and major axis of conic C respectively, and the product of eccentricity of hyperbola and conic C is 1, then -

(A) equation of hyperbola is $\frac{x^2}{36} - \frac{y^2}{12} = -1$.

(B) equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{3} = -1$.

(C) focus of hyperbola is $(0, 2\sqrt{3})$

(D) focus of hyperbola is $(0, -4\sqrt{3})$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q16. Match the properties given in **column-I** with the corresponding curves given in the **column-II**.

Column-I	**Column-II**
(A) The curve such that product of the distances of any of its tangent from two given points is constant, can be	(P) Circle
(B) A curve for which the length of the subnormal at any of its point is equal to 2 and the curve passes through (1, 2), can be	(Q) Parabola
(C) A curve passes through (1, 4) and is such that the segment joining any point P on the curve and the point of intersection of the normal at P with the x-axis is bisected by the y-axis. The curve can be	(R) Ellipse
(D) A curve passes through (1, 2) is such that the length of the normal at any of its point is equal to 2. The curve can be	(S) Hyperbola

Q17.

Column	**Column-II**
(A) The number of real common tangents to the circle $5x^2 + 5y^2 = 16$ and the hyperbola $3x^2 – y^2 = 48$, is	(P) 1
(B) The number of real common normals to parabola $y^2 = 4x$ and the circle $(x – 1)^2 + (y – 1)^2 = 1$, is	(Q) 2
(C) If P is any point on ellipse $\frac{x^2}{4}+4y^2=1$ whose foci are at A and B, then the maximum value of (PA) (PB) equals	(R) 3
(D) The length of latus-rectum of the parabola defined parametrically be equations $x = \cos t – \sin t$ and $y = \sin 2t$ equals	(S) 4
(E) The value of a for the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a > b)$, if the extremities of the latus-rectum of the ellipse having positive ordinate lies on the parabola $x^2 = –2(y – 2)$, is	(T) 5

Q18.

Column-I	Column-II
(A) If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, the locus of P is	(P) Straight line
(B) A variable circle C has the equation $x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$, where t is a parameter. The locus of the centre of the circle is	(Q) Circle
(C) The locus of point of intersection of tangents to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at two points the sum of whose eccentric angles is constant is	(R) Parabola
(D) An ellipse slides between two perpendicular straight lines. Then the locus of its centre is	(S) Hyperbola

SUBJECTIVE TYPE

Q19. Find the locus of the mid points of the chords of the circle $x^2 + y^2 = 16$, which are tangent to the hyperbola $9x^2 - 16y^2 = 144$.

Q20. Tangents are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$. Find the locus of mid-point of the chord of contact.

Q21. Chords of the hyperbola $x^2 - y^2 = a^2$ touch the parbola $y^2 = 4ax$. Prove that the locus of their mid points is the curve $(x - a)y^2 = x^3$.

Q22. If two points P and Q on the hyperbola $x^2/a^2 - y^2/b^2 = 1$ whose centre is C be such that CP is perpendicular to CQ and $a < b$, then prove that $(1/CP^2) + (1/CQ^2) = (1/a^2) - (1/b^2)$.

Q23. If the normal at a point P to the hyperbola $x^2/a^2 - y^2/b^2 = 1$ meets the x-axis at G, show that SG = e.SP, S being the focus of the hyperbola.

Q24. The perpendicular from the centre upon the normal on any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets at R. Find the locus of R.

Q25. A point P divides the focal length of the hyperbola $9x^2 - 16y^2 = 144$ in the ratio $S'P : PS = 2 : 3$ where S & S′ are the foci of the hyperbola. Through P a straight line is drawn at an angle of 135° to the axis OX. Find the points of intersection of this line with the asymptotes of the hyperbola.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	C	C	B	C	B	D	D	C	A
11	12	13	14	15	16	17	18	19	20
C	B	B	C	C	B	A	A	A	A
21	22	23	24	25	26	27	28	29	30
D	A	B	A	B	D	D	D	C	B
31	32								
A	C								

Ex - 1A

1. B,D **2.** A, B, C, D **3.** A, B, C, D **4.** A, B **5.** D
6. A **7.** A, B, C, D **8.** D **9.** C **10.** D
11. A → P ; B → Q ; C → P ; D → S **12.** $3x^2 - y^2 = 3a^2$
13. $7x^2 - 2y^2 + 12xy - 2x + 9y - 22 = 0$
14. (A) (–4,–1) ; (B) $\frac{5}{4}$; (C) (1,–1), (–9,–1); (D) $5x + 4 = 0$, $5x + 36 = 0$,
(E) $\frac{9}{2}$; (F) $(x + 4)^2 + (y + 1)^2 = 16$; (G) $(x + 4)^2 + (y + 1)^2 = 7$
15. $y = 2x \pm 1$ **16.** $\sqrt{2}$

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	A	B	B	A	C	D	B	C
11	12	13	14	15	16	17	18	19	20
A	D	B	A	A	A	B	C	A	A
21	22								
B	B								

Ex - 2A

1. B **2.** A,C,D **3.** B **4.** B,C **5.** A,B,C
6. B **7.** A,B,C,D **8.** C,D **9.** A,B **10.** B
11. D **12.** C **13.** A,B,C,D **14.** A,B,C,D **15.** A,D
16. (A)→(R), (S); (B)→(Q); (C)→(R); (D)→(P)
17. (A)→S; (B)→P; (C) →S; (D)→ P; (E)→Q
18. (A) S, (B) R, (C) P, (D) Q
19. $(x^2 + y^2)^2 = 16x^2 - 9y^2$.
20. $\frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9}\right)^2$ **21.** $x^3 = (x - a)$ **22.** not req. **23.** not req
24. $(x^2 + y^2)^2 (a^2y^2 - b^2x^2) = (a^2 + b^2)^2 x^2y^2$ **25.** $\left(\frac{-4}{7}, \frac{-3}{7}\right)$ & (–4, 3)